PRASAD LAB FOR MATERIALS RESEARCH

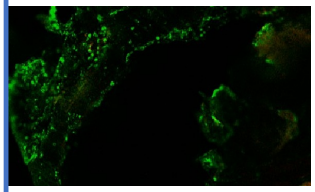
We use the mechanics of materials to build regenerative scaffolds for bone and heart repair, design multifunctional composites for defense and electronics, and advance plant mechanics and precision agriculture, combining nanomechanical experiments, manufacturing, and multiscale simulation.

Regenerative engineering

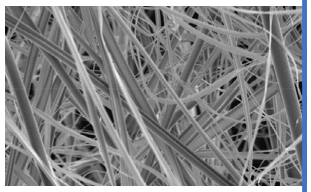
Scaffolds and devices for bone and heart repair



Open-source bioreactor for loading bone tissue



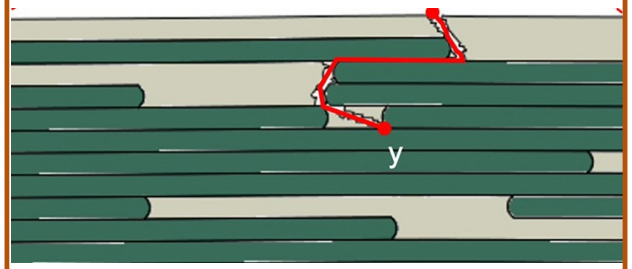
Live/dead staining of cultured bone



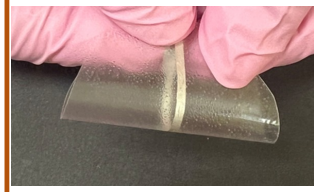
Electrospun cellulose fiber scaffold

Multifunctional composites

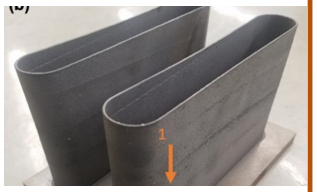
New materials for defense and electronics



Crack path in a MXene–polymer nanocomposite



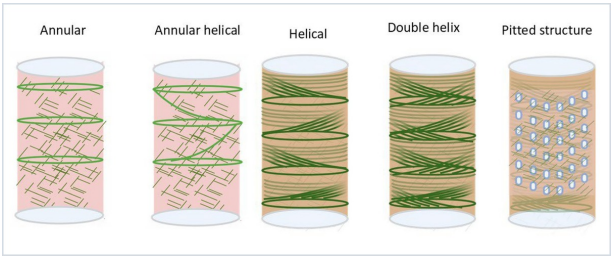
Flexible electronic substrate



3D-printed GRCop-42 alloy for rocket engines

Plant mechanics/Precision agriculture

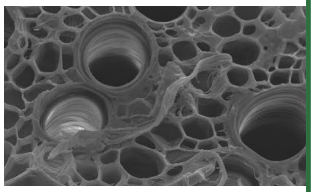
Cell wall mechanics and sensors for plant health



Cell wall patterning in growing xylem vessels



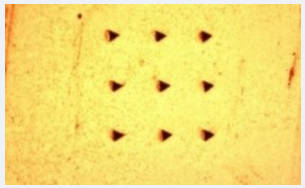
Wearable stem growth sensor

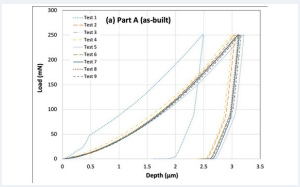


Xylem tissue under the electron microscope

Mechanics and Manufacturing Toolkit Snapshot

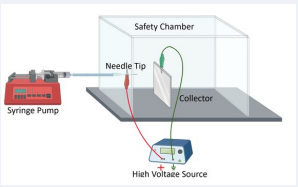
Nanomechanical experiments

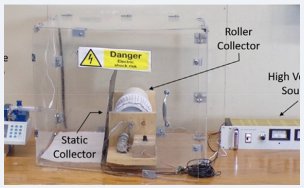




Indent arrays and load–depth curves

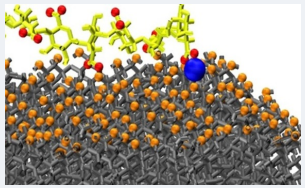
Manufacturing

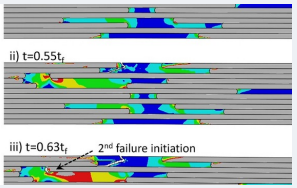




Lab-built electrospinning for fiber scaffolds

Multiscale simulation





Molecular dynamics to finite element models

Figures adapted from Prasad Lab publications, 2020–2026

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